

PHYSICS 101 · WEEK 1 QUIZ

Physics 101 — Week 1

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. What is pressure?

- ☐ A. How fast air moves from one place to another when it is squeezed.
- ☐ B. The amount of air that can be stored inside a closed container.
- ☐ C. The total weight of something pressing down, no matter how big or small it is.
- ☐ D. A push spread over an area — the same push on a smaller area makes more pressure.

2. Why does a thumbtack go into a corkboard when you push it, but your thumb alone doesn't?

- ☐ A. Metal is pulled toward cork the way iron is pulled toward a magnet.
- ☐ B. The tack is made of metal, which is heavier than skin, so it sinks into the cork.
- ☐ C. The tack's point puts your whole push onto a tiny area, so the pressure there is enormous.
- ☐ D. The tack pushes back against your thumb, and that push drives it into the board.

3. Blood pressure is written in mmHg. What does that unit mean?

- ☐ A. How many millimeters wide the artery is when the cuff squeezes it.
- ☐ B. How many milligrams of mercury are dissolved in the blood.
- ☐ C. How many times per minute the heart pushes blood into the arteries.
- ☐ D. How many millimeters high the pressure could push up a column of liquid mercury.

4. In 1643, Evangelista Torricelli filled a tube with mercury and stood it upside down in a dish. The mercury sank until a column about 760 mm tall remained. What did this show?

- ☐ A. That air has weight and presses down hard enough to hold up 760 mm of mercury.
- ☐ B. That the heart pumps blood into the arteries at a steady pressure.
- ☐ C. That liquids always sink to the same level no matter what tube they are in.
- ☐ D. That mercury boils at a much lower temperature than water does.

5. CPAP pressure is measured in cmH_2O . What does “6 cmH_2O ” mean?

- ☐ A. The water in the humidifier is kept at 6 degrees.
- ☐ B. A push strong enough to hold up a column of water 6 centimeters high.
- ☐ C. Six cubic centimeters of water flow into the tube with every breath.
- ☐ D. There are 6 centimeters of tubing between the machine and the baby's nose.

6. Which is the BIGGER pressure: 40 mmHg or 40 cmH_2O ?

- ☐ A. They are equal, because the number in front is the same.
- ☐ B. Neither – mmHg and cmH_2O measure two different things.
- ☐ C. 40 mmHg – mercury is much heavier than water, so lifting it takes more push.
- ☐ D. 40 cmH_2O – centimeters are bigger than millimeters, so it is the bigger unit.

7. About how big is the air pressure of the atmosphere at sea level?

- ☐ A. About 7,600 mmHg
- ☐ B. About 120 mmHg
- ☐ C. About 76 mmHg
- ☐ D. About 760 mmHg

8. Why do NICU teams use the gentlest ventilator pressure that works?

- ☐ A. Lower pressures keep the alarms quieter for the other babies in the unit.
- ☐ B. Too much pressure overstretches a preemie's delicate lungs and can even pop an air sac.
- ☐ C. Higher pressures use more electricity and can make the machine overheat.
- ☐ D. High pressure would push oxygen out through the baby's thin skin.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)